

HCI -Class Notes

Unit 1 Foundations of Human-Computer Interaction

Section 1: Human Capabilities

Give an overview of the study of HCI and its relevance to cognitive psychology and human physiology.

Human-computer interaction (HCI) is a **multidisciplinary** field in which psychology and other social sciences unite with computer science and related technical fields with the goal of making computing systems that are both **useful and usable**. It is a blend of applied and basic research, both drawing from psychological research and contributing new ideas to it. New technologies continuously challenge HCI researchers with new options, as do the demands of new audiences and uses. A variety of usability methods have been developed that draw upon psychological principles. HCI research has expanded beyond its roots in the cognitive processes of individual users to include social and organizational processes involved in computer usage in real environments as well as the use of computers in collaboration

What are the components of Information systems?

Inputs, outputs, feedback loops, and processes. Each = “black box”—the outside world cannot see what goes on inside it.

What are the components of Human cognitive system?

Storing, organizing, and processing information and making it available for later reference and use.

Both systems above works the same way.

What is the most successful application of the information-processing approach?

The study of human memory

What is generic memory model?

Memory is broken down into a sensory memory buffer, a short-term working memory, and a long-term memory, input is held briefly in the sensory buffer and is then loaded into working memory before being stored in long-term memory, input may also come into working memory from long-term memory

What is interaction design?

Designing interactive products to support the way people communicate and interact in their everyday and working lives.

What is the goal of interaction design?

- Develop usable products e.g. easy to learn, use, and provide enjoyable experience.
- Involve users in the design

What are the five principles of the EC Directive 90/270/EEC?

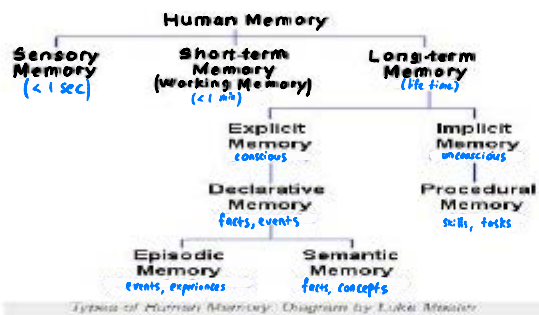
It's a legislation that requires employers to ensure the following when designing, selecting, commissioning or modifying software :

- That is suitable for the task
- That is easy to use and, where appropriate, adaptable to the user knowledge and experience.
- That it provides feedback on performance.
- That it displays information in a format and at a pace that is adapted to the user.
- That it conforms to the principles of software ergonomic.

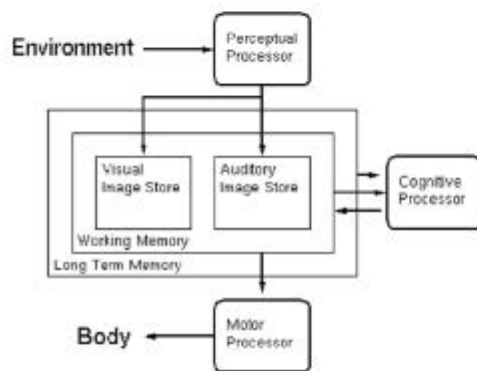
List and briefly describe the three functions of visual perception that are relevant to HCI

- **Perceiving size in depth:** We can identify similar objects regardless of the fact that they appear to us to be of vastly different sizes. In fact, we can use this information to judge distances.
- **Perceiving color:**
- **Perceiving brightness:** perception of brightness. Brightness is in fact a subjective reaction to levels of light. It is affected by luminance which is the amount of light emitted by an object. The luminance of an object is dependent on the amount of light falling on the object's surface and its reflective properties

Outline the three main types of human memory.



Describe the characteristics of the three stores of human memory.



Explain the possible relationship between short-term memory and long-term memory

- Information from short term memory is **stored** in long term memory **by rehearsal.**

Explain the information-processing approach to human reasoning and thinking.

It presents a way of thinking about human cognition and modeling aspects of it. Gestalt principles determine how people perceive information in the real world—on paper and on the computer screen.

Describe three types of reasoning

- Deductive reasoning: conclusion guaranteed - starts with the assertion of a general rule and proceeds from there to a guaranteed specific conclusion
- Inductive reasoning: conclusion merely likely - Inductive reasoning begins with observations that are specific and limited in scope, and proceeds to a generalized conclusion that is likely, but not certain, in light of accumulated evidence. You could say that inductive reasoning moves from the specific to the general

- Abductive reasoning: taking your best shot - typically begins with an incomplete set of observations and proceeds to the likeliest possible explanation for the set. Abductive reasoning yields the kind of daily decision-making that does its best with the information at hand, which often is incomplete.

Describe two long-term and two short-term individual differences relevant for HCI

- Long term- Sex, physical and intellectual abilities
- Short term - Effects of stress and fatigue

Provide an example of a computer system that would be greatly affected by short-term differences (open question).

- System that did not take into account the users who are under pressure, feeling ill or fatigue.

Provide an example of a computer system that would be greatly affected by long-term differences (open question).

- System that did not take into account the users that have some limitations i.e. visually impaired

Describe the role of psychology in computer interface design.

Principals and result from research of psychology has been distilled into guidelines for design, models to support design and techniques for evaluating design.

Describe three ways that psychology contributes to HCI.

By providing an in depth understanding of human:

- o information processing
- o capabilities
- o limitations
- o cooperative working
- o performance prediction

Section 2: The Computer

Give an overview of the configuration of a computer system.

A computer system comprises various elements each of which affect the

user of the system

Describe two broad categories of interaction devices

- Input devices : Allowing text entry, drawing and selection from screen
- Output Display devices allowing for interactive use.

Describe the three broad categories of text entry devices

- Alphanumeric keyboard : QWERTY keyboard
- CHORD Keyboards: Few keys 4 or 5, Microwriter
- Phone pad and T9 (Algorithm) entry: Dictionary guess

List nine 2D pointing devices

Mouse, touchpad, trackball, joystick

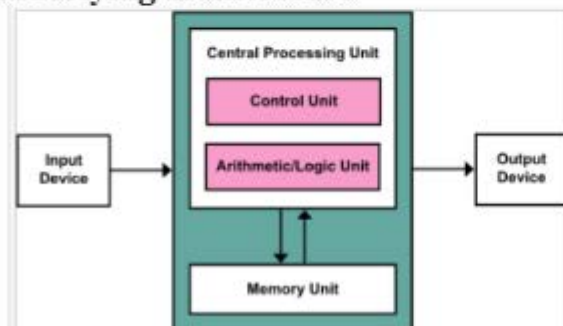
What is situated display? Give an example.

Displays in public places, typically screens only e.g. display screen in doctor's office.

Outline the problems with preparing a presentation for both print and screen

- Resolution, paper and screen has very different resolution
- Color (Most screens use additive color model), printers user subtractive color model.
- Fonts - Different width on screen and paper

Explain the Von Neumann bottleneck and specify the constraints of the underlying architecture



Von Neumann architecture scheme.



- an instruction fetch and a data operation cannot occur at the same time because they share a common bus
- limits the performance of the system

Describe four limitations on interactive performance presented in your textbook.

- Computation bound : Computation takes ages, causing frustration for the user.
- Storage channel bound: Bottleneck in transference of data from disk to memory
- Graphics bound: Updating displays requires a lot of effort
- Network Capacity: performance can be reduced by network speed.

Section 3: The Interaction

What are the extremes of interactivity in computer systems?

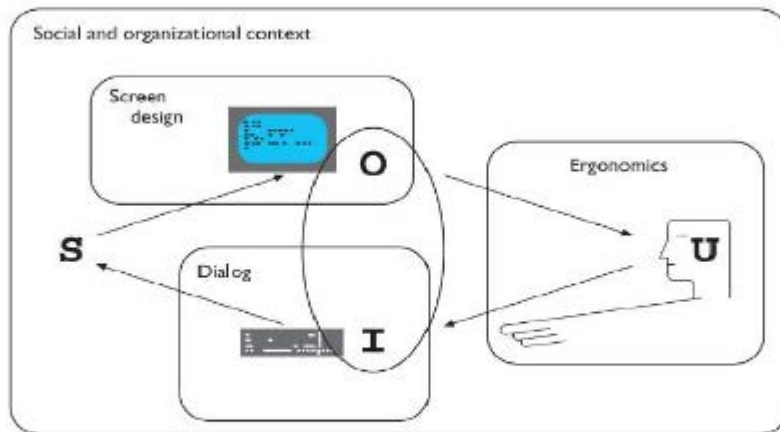
- Batch input - minimal interactions, user provides all information and let computer do its job.
- Direct manipulation and Virtual Reality: User directly providing instruction and getting feedback.

Describe the interaction framework outlined in your text

It attempts more realistic description of interaction by including the system explicitly and break it into 4 categories:

- System:
- User
- Input
- Output

Present the ACM framework for HCI



Define ergonomics and elaborate on its relationship to HCI.

Physical aspects of interfaces, it allows or define standards and guidelines for constraining the way we design certain aspects of system.

What are five human health issues for HCI to consider?

Physical position, environmental conditions (temperature, humidity), lighting, noise.

Describe five common interaction styles.

- Command line
- Menus
- Natural languages
- Question/answer
- Form fills
- WIMP
- POINT and click

List the defining characteristics of the WIMP interface.

Windows, icons, mice, and pull down.

Describe a WIMP interface that you use (open question).

MAC os/Windows OS

Describe the role of pre-emptive components in an interface oriented towards user initialization

It can act in a following manner:

- Block the application only while you can perform other tasks
- Block the entire web page e.g. MODAL

Annoying example: Dialog box ask user to confirm but the button to click ok is hidden.

Explain how context is important for the interaction between the human and the computer.

users work within a wider social and organizational context. This provides the wider context for the interaction, and may influence the activity and motivation of the user.

How can the design of the human-computer interface affect motivation?

Beautiful and/or novel interfaces are artistically pleasing and capable of fulfilling the tasks required – a marriage of art and science into a successful whole.

Why do some researchers add the criterion *satisfaction* to their consideration of usability? (p. 156 of *HCI*)

It is influenced by the user's perception of the quality of the work done, which leads to job satisfaction.

Provide a summary of the steps in an interaction.

Interaction Design involves four basic activities:

Identifying needs and establishing requirements

Developing alternative designs that meet those requirements

Building interactive versions of the designs so that they can be communicated and assessed

Evaluating what is being built throughout the process

Evaluating what has been built is the heart of Interaction Design

Section 4: Paradigms

Explain the meaning of the term *paradigm* with respect to interactive design.

New computing technologies arrive, creating a new perception of the

human computer relationship. **What are the two open questions for designers of interactive systems?**

How can an interactive system be developed to ensure its usability?. How can this usability be demonstrated or measured?

Unit 2-The design process

Section 1-Interaction design basic

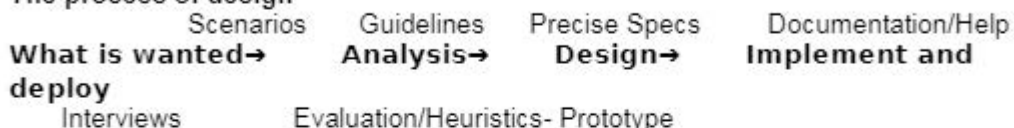
What is design?

- **Achieving goals within constraints**
 - goals - purpose who is it for, why do they want it
 - Constraints-What materials to use, platforms,time
 - trade-offs-Choosing between goals or constraints

Golden rule of design

- **Understanding your materials (Human and Computer)**
- understand computers
limitations, capacities, tools, platforms
- understand people
psychological, social aspects
- human error
and their interaction

The process of design



Agile Methods

Give one reason why iteration and prototyping are important in design. (p. 196 of HCI)

- Provides feedback which can improve the system.

1. **Iterations:** Iteration is not limited to the analysis—design—prototyping loop. It is considered beneficial to use iterative development to produce working, though limited, systems in order to receive quick feedback. Some variants of agile methods, for example extreme programming, suggest iterations of one or two weeks. This method results in a number of well-tested versions of the final system. As these versions are thoroughly tested, flaws such as bugs, misunderstanding of requirements, or changes of environment can be caught earlier.

2. **People oriented:** Rather than focus on rigorous design processes that all developers can follow, agile methods tend to focus more on the people involved, including, but not limited to, customers, developers, and testers.

Know your user

- who are they?
- probably not like you!
- talk to them
- watch them
- use your imagination

What is a persona?

- Description of an 'example' user (Not necessarily a real person)
- Describe the user as real with rich detail

What is a Scenario?

- Stories for design (describing situation and context)

What are three uses of scenarios?

1. Communicate with others (like clients and users)
2. Validate other models (played against task models)
3. Express Dynamics (What system looks like and how it behaves)

Navigation Design

- Local Structure= Single Screen (From one screen looking out)
- **What are the issues or concerns of local structure?**
 - knowing where you are
 - knowing what you can do
 - Knowing where you are going or what will happen
 - knowing where you've been or what you've done
- Global Structure- Whole Site (between screens within the application)
- **Levels**
 - widget choice-menus, buttons etc.
 - screen design
 - application navigation design
 - environment -other apps, O/S

What is the argument for and against deep hierarchy?

- They are difficult to navigate so it is better to use top-level categories

Screen Design and Layout

- Basic principles-Ask, think and design

- Tools for layout
 - grouping of items
 - order of items
 - decoration - fonts, boxes etc.
 - alignment of items
 - white space between items

Define affordance and give an example in which the affordance depends upon computer literacy?

- Things may suggest by their shape and other attributes what you can do with them eg button affords pushing (Users will click on a icon who are computer literate and will know to double click if single click doesn't work)

Iteration and prototyping

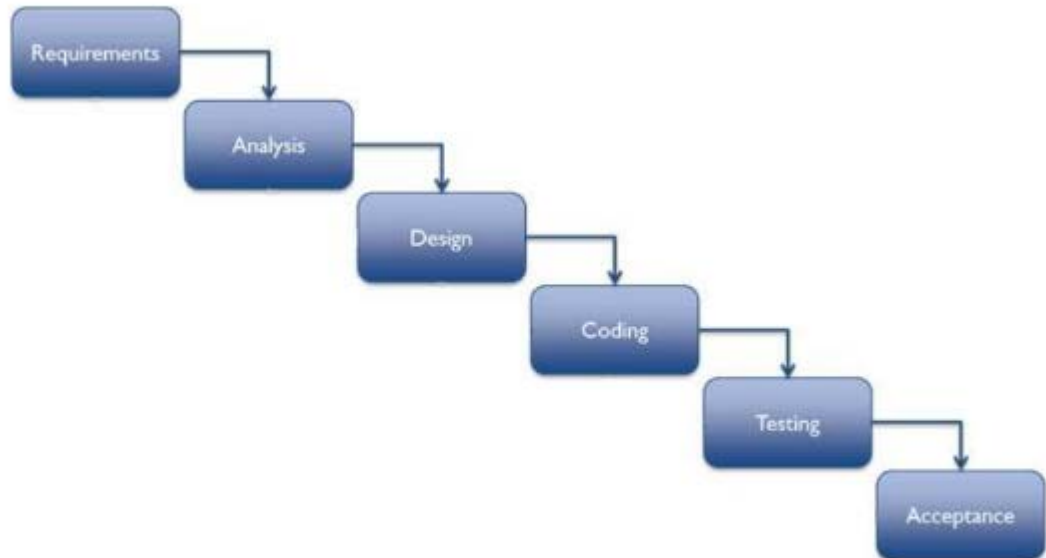
- First design is always not perfect and ideas need to be iterated

What are the two things that need to be done in order for prototyping to work?

1. To understand what is wrong and how to improve
2. A good start point

Section 2- HCI in the software process

What is the most important point to remember regarding HCI issues in the software design cycle? Relevant within all the activities of the software life cycle



Waterfall model

The waterfall model is a sequential (non-iterative) design process, used in software development processes, in which progress is seen as flowing steadily downwards (like a waterfall) through the phases of conception, initiation, analysis, design, construction, testing, production/implementation and maintenance.

Problems with traditional life cycles?

- Not highly interactive more like batch processing systems
- Does not promote the use of notations that support user perspective of the interactive system
- Models are not iterative

Usability engineering

- The ultimate test of usability based on measurement of user experience.
- Usability engineering demands that specific usability measures be made explicit as requirements

What is a problem in usability engineering?

- Usability metrics rely on measurements of very specific user actions in very specific situations
- Satisfies usability specs and not necessarily usability

Iterative design and prototyping

What are the three techniques for prototyping? Briefly describe the characteristics of each of them.

1. **Throw-away**-Throwing away prototypes to arrive at final requirements
2. **Incremental**-The final product is built as separate components, one at a time.
3. **Evolutionary**-Prototype is not discarded and serves as a basis for next design.

Problems with prototypes?

1. Time- building prototypes take time
2. Planning-Not enough experience for planning
3. non -functional features

Prototype techniques

- Storyboards-Graphical depiction of the appearance (animations)
- Limited functionality simulations- Designer can rapidly build graphical object and attach some behaviour to the objects

Problems with iterative designs?

- Initial prototypes have had bad features that will not be amended
- In the process of evaluation, a potential usability problem is diagnosed(important to understand problem not just the symptom.

Design Rationale

Design rationale is information that explains why a computer system is the way it is.

Benefits of design rationale

- communication throughout life cycle
- reuse of design knowledge across products
- enforces design discipline
- presents arguments for design trade-offs
- organizes potentially large design space
- capturing contextual information

What differentiates process-oriented design rationales from structure-oriented design rationales?

- Process-oriented: preserves order of deliberation and decision-making
- Structure-oriented design: emphasizes post hoc structuring of considered design alternatives
- Process-oriented design rationale is interested in recording an historically accurate description of design team making some decision on a particular issue for the design. While, structure-oriented design is less interested in preserving the historical evolution of design

Explain QOC design ?

- Questions, Options and Criteria notation. Designed by a set of questions representing the major issues of the design,

Section 3- Design Rules

List the key components of learnability, flexibility, and robustness.

1. **Learnability**- the ease with which new users can begin effective interaction and achieve maximal performance
 - a. Predictability-determining effect of future actions based on past interaction history
 - b. Synthesizability-assessing the effect of past actions(immediate vs. eventual honesty)
 - c. Familiarity-how prior knowledge applies to new system(guessability/affordance)
 - d. Generalizability-extending specific interaction knowledge to new situations
 - e. Consistency-likeness in input/output behaviour arising from similar situations or task objectives
2. **Flexibility**-the multiplicity of ways the user and system exchange information
 - a. Dialogue initiative-freedom from system imposed constraints on input dialogue system vs. user pre-emptiveness
 - b. Multithreading-ability of system to support user interaction for more than one task at a time concurrency vs. interleaving; multimodality
 - c. Task migratability-passing responsibility for task execution between user and system

- d. Substitutivity-allowing equivalent values of input and output to be substituted for each other
 - e. Customizability-modifiability of the user interface by user (adaptability) or system (adaptivity)
3. **Robustness**-the level of support provided the user in determining successful achievement and assessment of goal-directed behaviour
- a. Observability-ability of user to evaluate the internal state of the system from its perceivable representation
 - b. Recoverability-ability of user to take corrective action once an error has been recognized
 - c. Responsiveness-How the user perceives the rate of communication with the system
 - d. Task conformance-degree to which system services support all of the user's tasks

Standards

- set by national or international bodies to ensure compliance by a large community of designers standards require sound underlying theory and slowly changing technology
- **How does the applicability of standards differ for hardware and software?**
 - Standards for hardware are based on an understanding of physiology or human factors. On the other hand, software standards are based on theories from psychology or cognitive science

Guidelines

- more suggestive and general
- What is the purpose of a style guide?
 - Intended to help a programmer to capture the elements of the look and feel GUI. Promotes consistent between applications.

Shneiderman's 8 Golden Rules(important)

1. Strive for consistency
2. Enable frequent users to use shortcuts
3. Offer informative feedback
4. Design dialogs to yield closure
5. Offer error prevention and simple error handling
6. Permit easy reversal of actions
7. Support internal locus of control
8. Reduce short-term memory load

Norman's 7 Principles

1. Use both knowledge in the world and knowledge in the head.
2. Simplify the structure of tasks
3. Make things visible: bridge the gulfs of Execution and Evaluation.
4. Get the mappings right.
5. Exploit the power of constraints, both natural and artificial.
6. Design for error.
7. When all else fails, standardize.

Section 4- Universal Design

Universal Design- the process of designing products so that they can be used by as many people as possible in as many situations as possible. Eg disable people

Why should we not assume a "typical" user is like us?

- Because every user has different experience level than us.

7 universal design principles(important)

1. equitable use
2. flexibility in use
3. simple and intuitive to use
4. perceptible information
5. tolerance for error
6. low physical effort
7. size and space for approach and use

Multi-modal interaction

What is multi-modal interaction? What advantages might it have?

- Providing access to information through more than one mode of interaction
- Increases the bandwidth of the interaction between human and computer

Sound and Speech

What are three levels of speech in speech interaction?

1. Recording-Replaying messages
2. Transcription-Turning speech into text
3. Control-telling the computer what to do

What is an auditory icon?

- Use natural sounds to represent different types of objects and actions in the interface. Eg when you throw something in trash icon it makes the throwing away sound.

What is an earcon?

- Use structured combinations of notes called motives to represent actions and objective eg. beeping when error occurs on windows

Unit 4-Evaluation and User support

What is evaluation?

- tests usability and functionality of system
- should be considered at all stages in the design life cycle
- Three goals of evaluation
 - To assess the extent and accessibility of the system's functionality
 - To assess users experience of the interaction
 - To identify any specific problems

Cognitive walkthrough

- evaluates design on how well it supports user in learning task
- expert 'walks through' design to identify potential problems using psychological principles
- Four requirements for walkthrough
 - A spec or prototype of the system
 - A description of the task the user is to perform on the system
 - A complete list of actions to complete the task
 - A indication who the users are and there experience
- Four questions
 - Is the effect of the action the same as the user's goal at that point?
 - Will users see that the action is available ?
 - Willl they know what to do with action?
 - Will users understand feedback?

Nielsen's ten heuristics(important)

1. Visibility of system status

The system should always keep users informed about what is going on, through appropriate feedback within reasonable time.

2. Match between system and the real world

The system should speak the users' language, with words, phrases and concepts familiar to the user, rather than system-oriented terms. Follow real-world conventions, making information appear in a natural and logical order.

3. User control and freedom

Users often choose system functions by mistake and will need a clearly marked "emergency exit" to leave the unwanted state without having to go through an extended dialogue. Support undo and redo.

4. Consistency and standards

Users should not have to wonder whether different words, situations, or actions mean the same thing. Follow platform conventions.

5. Error prevention

Even better than good error messages is a careful design which prevents a problem from occurring in the first place.

6. Recognition rather than recall

Make objects, actions, and options visible. The user should not have to remember information from one part of the dialogue to another. Instructions for use of the system should be visible or easily retrievable whenever appropriate.

7. Flexibility and efficiency of use

Accelerators -- unseen by the novice user -- may often speed up the interaction for the expert user such that the system can cater to both inexperienced and experienced users. Allow users to tailor frequent actions.

8. Aesthetic and minimalist design

Dialogues should not contain information which is irrelevant or rarely needed. Every extra unit of information in a dialogue competes with the relevant units of information and diminishes their relative visibility.

9. Help users recognize, diagnose, and recover from errors

Error messages should be expressed in plain language (no codes), precisely indicate the problem, and constructively suggest a solution.

10. Help and documentation

Even though it is better if the system can be used without documentation, it may be necessary to provide help and documentation. Any such information should be easy to search, focused on the user's task, list concrete steps to be carried out, and not be too large.

Styles of evaluation with user

- Laboratory studies
 - Users take part in controlled tests in usability lab area.
 - Advantages: specialist equipment available and uninterrupted environment
 - Disadvantages: lack of context and difficult to observe several users cooperating
- Field study
 - Designer or evaluator taken out into user's work environment in order to observe the system in action

- o Advantages-natural environment and context retained (though observation may alter it) and longitudinal studies possible
- o Disadvantages:distractions and noise

Observational techniques (important)(pg 344)

1. Think aloud
2. Cooperative evaluation
3. Protocol analysis
4. Automated Analysis
5. Post-task walkthroughs

Query Techniques

1. Interviews
2. Questionnaire
 - a. General
 - b. Open ended
 - c. Scalar
 - d. Multi-choice
 - e. Ranked

Psychology methods (Measurement of feeling and how users react to the system)

1. Eye tracking(tracking the movement of eye with head set)
2. Psychology measurement (emotional response linked to physical changes)eg blood pressure and heart rate.

Section 2:User Support

Four types of support(important)

1. Quick reference(reminder to the user of the details of tools)
2. Task-specific help(help around a specific task)
3. Full explanation(full explanation of tool)
4. Tutorial (for new users-working through examples)

Requirements of good user support(important)

1. Availability-continuous access concurrent to main application
2. Accuracy and completeness-help matches and covers actual system behaviour
3. consistency-between different parts of the help system and paper documentation
4. Robustness-correct error handling and unpredictable behaviour
5. Flexibility-allows user to interact in a way appropriate to experience and task
6. Unobtrusiveness- does not prevent the user continuing with work

What are the five approaches to user support?

1. Command assistance (user requests help on command eg dos help)
2. Command prompts(error messages to help solve issue)
3. Context sensitive help(eg tooltips in excel)
4. Online tutorials
5. Online documentation(paper documents)
 - a. Use clear structure/heading

- b. Organize info according to user task
- c. Include FAQ
- d. User examples

Questions to ask when addressing help presentation?

- How is help requested?(command,button)
- How help is displayed?(new window)

Adaptive help-Domain specific expert systems and intelligent tutoring systems.Works by monitoring activity of user and constructing model of him.

Unit 5-User Models

Section 1:Cognitive Models

- Cognitive model is about the user understanding,processing and knowledge.

What is the difference between competence and a performance model?

- Competence models tend to be ones that can predict legal behaviour sequences but generally do this without reference to if user can execute or not.But performance model describe the necessary behaviour sequences and describe what what user need to execute the task.

Hierarchical representation of user task(actions) and goal(intentions)

- Mental process in which the user achieves goals by solving subgoal in divide and conquer way.
- GOMS(Goals Operators Methods Selection)
 - Goal-What user wants to achieve
 - Operator- basic actions user perform
 - Methods-decomposition of goals into subgoals and actions
 - Selection-means of choosing between competing methods
- CCT(Cognitive complexity theory)
 - Two parallel descriptions-one for user goals and the other for computer system
 - Uses production rules like If condition then action
- Problems with these methods?
 - A post hoc technique(defined by computer not sure)and expert vs novice
 - Good-can be used to express interface issue

Linguistic and grammatical models

- Understanding user behaviour and cognitive difficulty based on analysis of language between system and user
- BNF(Backus-Naur Form)
 - Syntactic level ignoring the semantics of the language
- TAG(Task-Action Grammar)
 - Parameterized grammar rules to emphasize consistency
 - Uses:User's existing knowledge/derived rules
 - Problems with goal and grammar based techniques?
 - Ignore system output-What users see

Physical and device-level models

- Keystroke-level model
 - Based on empirical knowledge of human motor system
 - Used to provide rough measures of user performance in terms of execution times for basic sequences of actions
 - Six execution phase motors
 - Physical- keystroking, pointing, homing and drawing
 - Mental preparation
 - System response
- Buxton's 3-state model
 - All input devices are not created equal in their capabilities, nor input techniques in their demands.
- Problem space model

Section 2: Socio-Organizational issues and stakeholder requirements

Organizational issues with information systems

- Conflict and power
 - Conflicting goals
 - System assuming cooperation will fail
 - Groupware effects organizational structures
- Invisible worker
 - Technology allows Working from distance can all which have economic benefits but management by presence doesn't work
- **What is the Free rider problem and how to solve?**
 - Not contributing equally and possibility of getting benefits without doing work
 - Solution: Increasing the visibility of participants contribution like an activity meter. Everyone has to contribute something
 - Critical mass-When the number of subscribers increases beyond the critical mass than the benefits for all dominate the cost.
- Benefits
 - Email or electronic conference help social contact

Capturing requirements

- **Who is the stakeholder?**
 - Anyone who is affected by the success or failure of the system like customers, managers, shareholder and competitors.
- **Who are the stakeholder in CUSTOM approach**
 - Primary-end users (eg new booking system for travel agents will have travel agents and staff as primary stakeholders)
 - Secondary-people who do not directly use the system (someone who receiver's report produced by system)-Customers, airline mgmt
 - Tertiary-someone who doesn't fall in primary or secondary but is affected directly like the director-competitors and airline shareholders
 - Facilitating -people who are involved in the design, development and maintenance of the system-design team and IT

- Socio-technical models
 - Response to technological determinism
 - Describes impact of specific technology on organization
 - Several approaches
 - **CUSTOM**-Socio-technical methodology designed to be practiced in small organizations. Based on User skills and task match(USTM) to identify stakeholder requirements.
 - Six stages
 - Describe org context including primary goals
 - Identify and describe stakeholders
 - Identify and describe work groups
 - Identify and describe task-object pairs
 - Identify stakeholder needs
 - Consolidate and check stakeholder requirements
 - **OSTA-Open system task analysis**
 - Describes what happens when a technical system is introduced into an organizational work environment. The difference between CUSTOM and OSTA is the CUSTOM focused more on stakeholder requirements while OSTA focuses on tasks
 - Eight main stages
 - primary task identified in terms of users' goals
 - task inputs to system are identified
 - external environment into which the system will be introduced is described, including physical, economic and political aspects
 - transformation processes within the system are described in terms of actions performed on or with objects
 - social system is analyzed, considering existing internal and external work-groups and relationships
 - technical system is described in terms of configuration and integration with other systems
 - performance satisfaction criteria are established, indicating social and technical requirements of system
 - new technical system is specified
- **Soft Systems methodology**
 - **How is success defined by soft systems?**
 - no assumption of technological solution - emphasis on understanding situation of the context of technological developments and the influences and concerns that exist within the system under consideration.
 - seven stages recognition of problem and initiation of analysis
 - detailed description of problem situation-rich picture

- generate root definitions of system-
CATWOE(client,actor,transformation,weltanschauung(
how system is perceived),owner and environment)
- conceptual model - identifying transformations
- compare real world to conceptual model
- identify necessary changes
- determine actions to effect changes
- **Participatory design-**
 - Philosophy that encompasses the whole design cycle
 - Eg storyboards,workshops
- **ETHICS(Effective technical and human implementation of computer -based systems)**
 - Stakeholder included as participants in the decision making
- **Ethnographic methods**
- **Contextual inquiry**

Unit 6 Task Models and Dialogs

Section 1:Analyzing Tasks

Task Analysis-Methods to analyse people's jobs and how they perform it. Recognize the importance of user and look more at observable behaviour users.Comes in requirements in life cycle.

Three approaches to task analysis

1. **Task decomposition-** task split into subtasks
2. **Knowledge-based techniques-**what user needs to know about the action involved in task
3. **Entity-relation based analysis-** identifying actions and objects and the relationship between them

Hierarchy task Analysis (HTA)

- Fixed sequence(1.1 then 1.2)
- Optional tasks
- Wait for events
- Cycles
- Time-sharing (same time tasks)
- Discretionary (do any)
- mixtures(several of above)

Example

0.in order to clean house

1. Get the vacuum
2. Get attachments
3. Clean the rooms
 - 3.1 clean the hall
 - 3.2 clean living room
4. Empty dust bag

5. Put away

Knowledge-based techniques

- Begins by listing all the objects and actions involved in the action and then building taxonomies (like animal hierarchical description) of them
- Eg Motor controls
 - Steering-Steering wheel, indicators
 - Engine
 - Direct-acceleration
 - gearing -gear stick

What is TAKD?

- Task analysis for knowledge description uses a special form of taxonomy called task descriptive hierarchy. Either or branches.
- Three types of branches
 - a. XOR-Normal-Object in one and only one branch
 - b. AND-Object must be in both
 - c. OR-weakest case-can be in one or many or none.

Entity Relationship Techniques

- identifying actions and objects and the relationship between them like in database
- Start with list of objects and classify them
 - Actors and attributes
 - Agent-who performs actions
 - Three relationships
 - Object-Object
 - Actions-Events
 - Action-Object

What are three sources of information?

1. Documentation-Manuals
2. Interviews
3. Observations

Three Uses of task analysis?

1. The production of manuals and tuition material
2. Requirements capture and high-level systems design
3. Detailed interface design

Section 2: Dialogue notation and design

Dialog

- Conversion between two or more parties
- Three levels of computer language
 - Lexical-Lowest level (the shape of icons and keys pressed)-Sound/spelling of words
 - Syntactic (inputs and outputs)-Grammar of sentence construction
 - Semantic-Meaning of conversion in terms of its effect on computer internal data structure-the meaning ascribed by different participants to the conversation.
- Why use separate **dialog description notation**

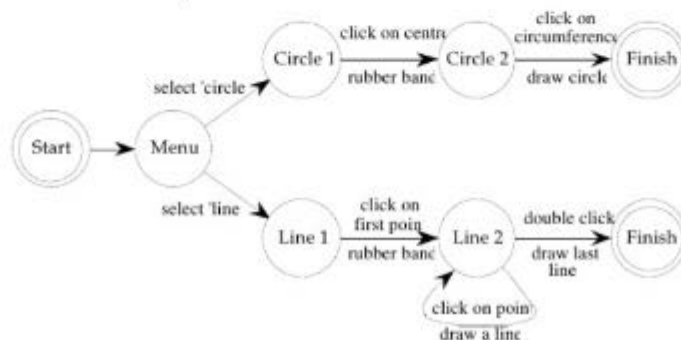
- Ease of analysis and separation of interface from actual calculations
- Write down dialogue before program is written

Diagrammatic notations

- State transition networks(STNS)

State transition networks (STN)

- circles - states
- arcs - actions/events



What is the difference between flowcharts and STNS?

- Boxes in flow chart represent process and decision
- Wide range of box shapes to represent activities
- Simple and useful

Unit 7 Groupware, Ubiquitous Computing, Virtual and augmented reality, Hypertext and multimedia

Section 1: Groupware and computer-supported collaborative work

How is CSCW(computer-supported cooperative work) related to HCI?

- Both draw on knowledge from a wide range of disciplines where the principle in HCI is psychology-computing and sociology-computing in CSCW

What is groupware?

- The provision of computer systems to support group working eg video conferencing
- Software designed for group work

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HUMAN-COMPUTER
INTERACTION

The Time/Space Matrix

Classify groupware by:

when the participants are working,
at the same *time* or not

where the participants are working,
at the same *place* or not

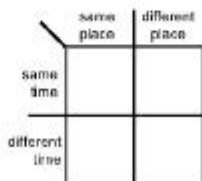
Common names for axes:

time:

synchronous/asynchronous

place:

co-located/remote



There types of groupware

1. Computer-mediated communication(CMC)
 - Direct communication
 - Email and bulletin boards
 - Structured message systems-Filtered form of email
 - Instant messaging and SMS
 - Video conference and virtual space
2. Meeting and decision support systems(sharing of idea and recording)
 - a. Argumentation tools-record the arguments used to arrive at a decision
 - b. Meeting rooms-face to face groups synchronous co-located in brainstorming and management meetings.Shared screens
 - c. Shared drawing surfaces-can be used for synchronous design meetings
 - i. White board
 - ii. LCD projection on whiteboard
3. Shared application and artifacts (sharing of work domains like computer sharing)
 - a. Shared PCs and shared window systems-Two or more computer which function as if they are one like screen sharing
 - b. Shared editors -Collaboration aware :one document several users like google docs
 - c. Co-authoring systems(long term document production not for editing)
 - i. Shared text editing is short-term while co-authoring is long term
 - ii. Shared editing is synchronous and co authoring is asynchronous
 - d. Shared diaries (meeting calendars shared)

Frameworks for groupware

- **Granularity of sharing**
 - Chunk size and update frequency
- **What are the five ways of testing for robustness?**
 - Rebooting system
 - Kill client process
 - Random input
 - Break the system

Section 2: Ubiquitous Computing

Ubiquitous computing

- Any computing technology that permits human interaction away from a single workstation eg pen based technology and wireless networks
- **What does implicit input mean?** Our natural interactions with the physical environment provide sufficient input to a variety of attendant service without any further user interaction. Sensor based input

Virtual reality has found a number of applications in the games market. Is this a suitable use of such technology? Discuss the possible benefits and disadvantages of exploiting leading-edge technology in a leisure market.

answer

The leisure market is important to the computer industry and, being commercially lucrative, can provide an impetus for development which can then be exploited in other spheres.

If virtual reality takes off in games it will result in the development of cheaper equipment and more robust techniques, since the games customer tends to be young (and not overly rich) but with sophisticated expectations. Indeed, many research projects in universities are currently using equipment developed specifically for games, since the more powerful, general purpose equipment is prohibitively expensive.

Another possible advantage is that it makes the technology familiar and therefore accessible and acceptable.

A possible disadvantage is that the use of this technology in games may trivialize it so that it is not considered as a serious solution to other more weighty problems. However, this seems less likely than the advantages outlined above

Context Aware computing

- Making interactions with services more seamless and less distracting from our everyday activities.
- Ability to easily capture and store memories of live experiences and serve them for later use

Automated capture and access

- The task of preserving a record of some live experience that then reviews at some point in the future

Toward continuous interaction

- Providing continuous interaction moves computing from a localized tool to a constant presence.

EXERCISE 21.3

What factors are likely to delay the widespread use of video in interfaces? What applications could benefit most from its use?

answer

Some of the factors are the costs in terms of hardware and software for compression and decompression; the slow speed due to the high bandwidth; the overall cost of equipment (for example, camera, video, CD); the lack of design tools to exploit video; the lack of specialist skills amongst designers. Many applications have been suggested as candidates for the integration of video. Educational systems, games and help systems are liable to benefit since information can be passed more clearly and memorably and new dimensions added. Other areas such as virtual reality can use video together with graphics in the creation of their artificial worlds. CSCW systems can use video to provide a face-to-face communication link between distributed workers (see Chapters 14 and 19). However, although these appear to be areas where video has a promising future, its use needs to be carefully considered and its consequences investigated. It may be that it will not fulfil its initial promise.

understanding hypertext escapes linearity, words and the page

- Hypertext media-Additional media such as video and photos
- Animations-Adding motion to images
- Audio issues-Formats and annoying if unwanted

finding things-navigating hyperspace

- Making navigation easier
 - Maps-give overview of structure show current location
 - Recommended routes-Guided tour
 - Levels of access
 - Supporting printing-links back to source

web technology-how it all works

- Web servers and clients sends information between browser and web server
- Network issues such as bandwidth time and reliability

web content

static: unchanging pictures and text

dynamic: interaction and applications on the web